

The First-Order Contrast Operator

The elastodynamic matrix–vector wave equation as a route to the single-site T -matrix: derivation, validation, and what it actually buys

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Abstract

The elastodynamic matrix–vector wave equation $\partial_3 \mathbf{q} = \mathcal{A} \mathbf{q} + \mathbf{d}$ expresses the wave field as a first-order recursion in depth. This note asks whether the cube moment method can be applied to it directly, to obtain a single-site T -matrix without passing through the Navier equation. The answer is yes, and the construction is not new: the thesis already carries it in 2 $\frac{1}{2}$ -D. What is new here is the 3-D Cartesian operator, and a measurement of what the route is worth.

Three results. *First*, the operator matrix $\mathcal{A}$ is derived from the stiffness tensor and validated against two independent objects that agree under one constant similarity transform; the thesis shorthand ζ, χ are shown to be the paper’s v_1, v_2 . *Second*, the test space is *derived* rather than chosen: $\mathbf{J}_6$ is antisymmetric and $\Delta \mathcal{A}^t \mathbf{J}_6 = -\mathbf{J}_6 \Delta \mathcal{A}$, so $\mathbf{J}_6 \Delta \mathcal{A}$ is symmetric and the only natural projection is test = $\mathbf{J}_6 \times$ trial. This dissolves an apparent mismatch of ten conditions against nine unknowns. *Third*, and least expected, the resulting coupling agrees with the ordinary Navier weak form at Born order but differs at $O(\Delta c^2)$ entirely within the traction channel — and that difference splits into a genuine O_h -invariant resummation *and* a spurious preference for the x_3 axis, in comparable amounts. A single-site score against an independent arbiter in a homogeneous background would mix the two and settle neither. The directional preference ceases to be an artefact once the background is layered, which is where the route was wanted.

Every claim below is backed by an executable check; the two scripts report 25/25 and 44/44, and a separate gate binds the numerical propagator to the derived $\mathcal{A}$. Section 10 states plainly what is *not* yet established.

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1 Conventions and objects

1.1 Indices, time, and units

The positive x_3 axis points downward; x_1, x_2 are lateral. Greek indices α, β run over the lateral pair $\{1, 2\}$ only, Latin indices over $\{1, 2, 3\}$. The time convention is $e^{-i\omega t}$ throughout, so $\partial_t \rightarrow -i\omega$ and the particle velocity is $\mathbf{v} = -i\omega \mathbf{u}$. The lateral transform is $e^{+i(k_1 x_1 + k_2 x_2)}$, hence $\partial_\alpha \rightarrow ik_\alpha$.

Numerical work uses seismic units: km, km s^{-1} , g cm^{-3} , GPa. These are consistent, since $\text{GPa} = \text{g cm}^{-3} \cdot (\text{km s}^{-1})^2$.

Two orderings appear in the sources and must not be conflated. The paper orders components $(1, 2, 3)$; the thesis orders them $(z, x, y) = (x_3, x_1, x_2)$. The map between them is written explicitly in §3.2.

1.2 The field vector

The matrix–vector wave equation is

$$\partial_3 \mathbf{q} = \mathcal{A} \mathbf{q} + \mathbf{d}, \quad \mathbf{q} = \begin{pmatrix} \mathbf{q}_1 \\ \mathbf{q}_2 \end{pmatrix}, \quad \mathcal{A} = \begin{pmatrix} \mathcal{A}_{11} & \mathcal{A}_{12} \\ \mathcal{A}_{21} & \mathcal{A}_{22} \end{pmatrix}, \quad (1)$$

with the elastodynamic choice

$$\mathbf{q}_1 = -\boldsymbol{\tau}_3, \quad \mathbf{q}_2 = \mathbf{v}, \quad (\boldsymbol{\tau}_j)_i = \tau_{ij}, \quad (2)$$

so that $\mathbf{q} = (-\tau_{13}, -\tau_{23}, -\tau_{33}, v_1, v_2, v_3)^t$. This is not an arbitrary pairing: $-\boldsymbol{\tau}_3$ and $\mathbf{v}$ are conjugate in the vertical power flux, $j = \frac{1}{4}(-\boldsymbol{\tau}_3^\dagger \mathbf{v} - \mathbf{v}^\dagger \boldsymbol{\tau}_3)$, and that is what makes the metric of §1.3 the natural one.

1.3 The metric $\mathbf{J}_6$

The symplectic metric is

$$\mathbf{J}_6 = \begin{pmatrix} \mathbf{0} & \mathbf{I}_3 \\ -\mathbf{I}_3 & \mathbf{0} \end{pmatrix}, \quad \mathbf{J}_6^t = -\mathbf{J}_6, \quad \mathbf{J}_6^2 = -\mathbf{I}_6. \quad (3)$$

A naming warning. This project, following the thesis, calls this matrix $\mathbf{J}_6$; it is already defined as such in the package. The paper calls the *same* matrix $\mathbf{N}$, and uses $\mathbf{J}$ for $\text{diag}(\mathbf{I}_3, -\mathbf{I}_3)$, a different object. Translate on the way in. Every occurrence of $\mathbf{N}$ in the paper’s equations (16)–(30) corresponds to $\mathbf{J}_6$ here.

The operator matrix is *quasi-Hamiltonian* with respect to it,

$$\mathcal{A}^t \mathbf{J}_6 = -\mathbf{J}_6 \mathcal{A}, \quad (4)$$

which is the paper’s eq. (27) and the thesis’s ATdef. Everything in §6 follows from (4) together with the antisymmetry of $\mathbf{J}_6$, and from nothing else.

2 The first-order system for an isotropic medium

2.1 From the stiffness tensor to the stiffness matrices

For an isotropic solid,

$$c_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad \rho_{ij} = \rho \delta_{ij}, \quad (5)$$

with $\lambda = \lambda(\mathbf{x}, \omega)$, $\mu = \mu(\mathbf{x}, \omega)$, $\rho = \rho(\mathbf{x}, \omega)$ and $K_c = \lambda + 2\mu$. The stiffness matrices are $(\mathbf{C}_{jl})_{ik} = c_{ijkl}$. In particular

$$\mathbf{C}_{33} = \text{diag}(\mu, \mu, K_c), \quad \mathbf{C}_{33}^{-1} = \text{diag}(\mu^{-1}, \mu^{-1}, K_c^{-1}), \quad (6)$$

and $\mathbf{C}_{jl}^t = \mathbf{C}_{lj}$, which follows from $c_{ijkl} = c_{klij}$. All nine blocks are reproduced from (5) and checked entry by entry.

2.2 Eliminating the in-plane tractions

The components $\boldsymbol{\tau}_1, \boldsymbol{\tau}_2$ do not appear in (2) and must be removed. Doing so introduces the reduced stiffness

$$\mathbf{U}_{\alpha l} = \mathbf{C}_{\alpha l} - \mathbf{C}_{\alpha 3} \mathbf{C}_{33}^{-1} \mathbf{C}_{3l}, \quad (7)$$

with the two structural properties $\mathbf{U}_{\alpha 3} = \mathbf{0}$ and $\mathbf{U}_{\alpha\beta}^t = \mathbf{U}_{\beta\alpha}$. Both are checked rather than assumed, because it is they — not the algebra that produced them — that make the symmetry relations of §3.3 hold.

The isotropic $\mathbf{U}_{\alpha\beta}$ is where two constants are *born*:

$$v_1 \equiv (\mathbf{U}_{11})_{11} = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad v_2 \equiv (\mathbf{U}_{12})_{12} = \frac{2\mu\lambda}{\lambda + 2\mu}, \quad v_1 - v_2 = 2\mu. \quad (8)$$

They are read off $\mathbf{U}$ and only then compared with the published definitions; they are not independent quantities. Every $\mathbf{U}_{\alpha\beta}$ has a null third row and column, which is the statement that τ_{33} has been fully eliminated.

2.3 The operator matrices

The general construction gives

$$\mathcal{A}_{11} = -\partial_\alpha \mathbf{C}_{\alpha 3} \mathbf{C}_{33}^{-1}, \quad \mathcal{A}_{12} = i\omega\rho - \frac{1}{i\omega} \partial_\alpha \mathbf{U}_{\alpha\beta} \partial_\beta, \quad \mathcal{A}_{21} = i\omega \mathbf{C}_{33}^{-1}, \quad \mathcal{A}_{22} = -\mathbf{C}_{33}^{-1} \mathbf{C}_{3\beta} \partial_\beta, \quad (9)$$

and for the isotropic medium these evaluate to

$$\mathcal{A}_{11} = \begin{pmatrix} 0 & 0 & -\partial_1 \frac{\lambda}{\lambda+2\mu} \\ 0 & 0 & -\partial_2 \frac{\lambda}{\lambda+2\mu} \\ -\partial_1 & -\partial_2 & 0 \end{pmatrix}, \quad \mathcal{A}_{22} = \begin{pmatrix} 0 & 0 & -\partial_1 \\ 0 & 0 & -\partial_2 \\ -\frac{\lambda}{\lambda+2\mu} \partial_1 & -\frac{\lambda}{\lambda+2\mu} \partial_2 & 0 \end{pmatrix}, \quad (10)$$

$$\mathcal{A}_{12} = \begin{pmatrix} i\omega\rho - \frac{1}{i\omega}(\partial_1 v_1 \partial_1 + \partial_2 v_2 \partial_2) & -\frac{1}{i\omega}(\partial_2 \mu \partial_1 + \partial_1 v_2 \partial_2) & 0 \\ -\frac{1}{i\omega}(\partial_2 v_2 \partial_1 + \partial_1 \mu \partial_2) & i\omega\rho - \frac{1}{i\omega}(\partial_1 \mu \partial_1 + \partial_2 v_1 \partial_2) & 0 \\ 0 & 0 & i\omega\rho \end{pmatrix}, \quad (11)$$

with $\mathcal{A}_{21} = i\omega \operatorname{diag}(\mu^{-1}, \mu^{-1}, K_c^{-1})$.

The ordering is the content. Compare the $(1, 3)$ entry of $\mathcal{A}_{11}$ with the $(3, 1)$ entry of $\mathcal{A}_{22}$. In the first the derivative stands to the *left* of the modulus and differentiates the product; in the second it stands to the *right* and differentiates the field alone. For a laterally invariant medium the distinction collapses and is invisible. For a laterally varying one it is the physics. This is why the checks in §3.1 compare *operators* — by their action on generic fields — rather than expressions, and why they are accompanied by a negative control that corrupts precisely this ordering.

3 Validating $\mathcal{A}$ against two independent objects

3.1 Arbiter A: the published isotropic operator matrices

The blocks (10)–(11) are transcribed separately and compared against those built from (9). The comparison is made by acting on a vector of generic symbolic functions, so it tests the derivative ordering as well as the coefficients. All four match.

A check that cannot fail is worth nothing, so a corrupted $\mathcal{A}_{22}$ — with the derivative wrongly placed outside the modulus — is also tested, and is rejected.

3.2 Arbiter B: the thesis system matrix

The thesis carries a 6×6 Fourier-domain system matrix $\mathbf{A}(k_x, k_y)$ for a laterally invariant medium, in a *different* field vector

$$\mathbf{b} = (u_3, u_1, u_2, \tau_{33}, \tau_{31}, \tau_{32})^t, \quad (12)$$

i.e. displacement–stress rather than velocity–traction, ordered (z, x, y) rather than $(1, 2, 3)$, and normalised so that $\rho\omega^2$ appears where the paper has $i\omega\rho$. It uses the shorthand

$$\gamma = \frac{\lambda}{\lambda + 2\mu}, \quad a = \frac{1}{\lambda + 2\mu}, \quad b = \frac{1}{\mu}, \quad \zeta = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad \chi = \frac{2\mu\lambda}{\lambda + 2\mu}. \quad (13)$$

Comparing (8) with these: $\zeta \equiv \nu_1$ and $\chi \equiv \nu_2$. The thesis’s in-plane stiffnesses and the paper’s auxiliary moduli are the same two quantities, born in the same elimination (7). This is verified, not asserted.

Setting λ, μ, ρ constant and acting on $e^{i(k_1 x_1 + k_2 x_2)}$ reduces $\mathcal{A}$ to a matrix. With the constant map $\mathbf{b} = \mathbf{S}\mathbf{q}$ built from $\mathbf{u} = \mathbf{v}/(-i\omega)$, $\boldsymbol{\tau}_3 = -\mathbf{q}_{1:3}$ and the permutation $(1, 2, 3) \rightarrow (3, 1, 2)$, the identity

$$\mathbf{A}_{\text{thesis}} = \mathbf{S} \mathbf{A}_{\text{paper}} \mathbf{S}^{-1} \quad (14)$$

holds entry by entry. Since the two matrices were derived independently and published separately, this is evidence rather than bookkeeping.

3.3 Structural checks

Three further properties are verified on the reduced 6×6 :

1. the operator symmetry relations $\mathcal{A}_{11}^t = -\mathcal{A}_{22}$, $\mathcal{A}_{12}^t = \mathcal{A}_{12}$, $\mathcal{A}_{21}^t = \mathcal{A}_{21}$, where the transpose is the formal adjoint under integration by parts ($\partial_\alpha^t = -\partial_\alpha$);
2. the quasi-Hamiltonian relation (4), which in the wavenumber domain reads $\mathbf{A}^t(-\mathbf{k})\mathbf{J}_6 = -\mathbf{J}_6\mathbf{A}(\mathbf{k})$;
3. the characteristic polynomial factorises as $(s^2 + k_{z,p}^2)(s^2 + k_{z,s}^2)^2$, recovering $\alpha^2 = (\lambda + 2\mu)/\rho$ and $\beta^2 = \mu/\rho$ from a construction in which no wave speed was ever written down.

Item 3 is the one that binds the algebra to physics: nothing upstream knows the wave speeds.

4 The contrast operator

4.1 Definition and the Lippmann–Schwinger equation

Inside the scatterer the medium is $(\lambda + \Delta\lambda, \mu + \Delta\mu, \rho + \Delta\rho)$. Writing $\mathcal{A}^{\text{tot}} = \mathcal{A} + \Delta\mathcal{A}$, the total field solves the background system driven by its own contrast,

$$(\partial_3 - \mathcal{A})\mathbf{q} = \Delta\mathcal{A}\mathbf{q} + \mathbf{d}, \quad (15)$$

which is the Lippmann–Schwinger equation of the first-order system.

4.2 The foundation check

Before (15) means anything, $\mathbf{q}$ must actually satisfy (1). Building $\mathbf{q}$ from a *generic* displacement field in a *laterally varying* medium and forming $\partial_3\mathbf{q} - \mathcal{A}\mathbf{q}$ gives two halves with different strengths:

- rows 4–6 vanish **identically**, for any $\mathbf{u}$ whatever. They are the constitutive relation rearranged; no equation of motion is used. Failure here would mean the elimination (7) is wrong.
- rows 1–3 equal **exactly** the Navier body force $f_i = -\rho\omega^2 u_i - \partial_j\tau_{ij}$. Anything left over would be a defect in $\mathcal{A}_{11}$ or $\mathcal{A}_{12}$.

Both hold. This is a considerably stronger statement than the plane-wave and eigenvalue checks of §3.3, because $\mathbf{u}$ is generic and the moduli vary laterally, so every ordering term in (10) is exercised.

4.3 $\Delta\mathcal{A}$ is not linear in the contrast

The Navier potential is exactly linear in $(\Delta\lambda, \Delta\mu, \Delta\rho)$. $\Delta\mathcal{A}$ is not, because the elimination inverted $\mathbf{C}_{33}$:

$$(\Delta\mathcal{A}_{21})_{11} = i\omega \left(\frac{1}{\mu + \Delta\mu} - \frac{1}{\mu} \right) = -\frac{i\omega \Delta\mu}{\mu(\mu + \Delta\mu)}, \quad (16)$$

whose ratio to its linearisation $-i\omega\Delta\mu/\mu^2$ is exactly $\mu/(\mu + \Delta\mu)$. At the project’s validated moderate contrast ($\alpha = 5$, $\beta = 3$, $\rho = 2.5$, so $\lambda = 17.5$, $\mu = 22.5$ GPa; $\Delta\lambda = 2$, $\Delta\mu = 1$ GPa):

quantity	value
$\Delta\mu/\mu$	0.04444
exact / linear for $\Delta b = \Delta(1/\mu)$	0.95745
exact / linear for $\Delta a = \Delta(1/K_c)$	0.93985

The first-order form has already summed part of the local response exactly. §7 measures what that is worth, and finds the answer is not the obvious one.

5 The augmentation

5.1 Why a derivative must be moved

Inside the scatterer the contrasts are constant, so every coefficient of $\Delta\mathcal{A}$ carries the indicator $\mathbf{1}_V$. A derivative standing to the *left* of such a coefficient differentiates the indicator and produces a surface delta on the cube faces. Integration by parts over all space moves that derivative onto the test function instead, where it meets something smooth:

$$\int \phi_i (-\partial_\alpha (m \mathbf{1}_V \psi_j)) d\mathbf{x} = + \int (\partial_\alpha \phi_i) m \mathbf{1}_V \psi_j d\mathbf{x}, \quad (17)$$

and the face term never arises. This is the step the thesis describes as transferring the derivatives onto the source and receiver coordinates of the propagator, and it is what converts a singular integro-differential equation into a nonsingular, compact integral equation.

5.2 The augmented dimension

After the transfer every term has the form $(\partial^a \phi)^t \cdot \text{const} \cdot (\partial^b \psi)$ with $a, b \in \{0, 1\}$, so the form is *multiplicative* on the enlarged vector $\mathcal{L}\psi = (\psi, \partial_1 \psi, \partial_2 \psi)$ — nominally $6 + 6 + 6 = 18$ slots. The true dimension is smaller and is counted, not assumed.

Result. Exactly $10 = 6 + 4$ slots are used. The four extra are $\partial_1 v_1, \partial_2 v_1, \partial_1 v_2, \partial_2 v_2$: the lateral gradients of the two *horizontal* velocity components, and nothing else.

Setting $\partial_2 \rightarrow ik_y$, as a $2\frac{1}{2}$ -D formulation does, makes two of those four multiplicative, leaving $6 + 2 = 8$. That is precisely the thesis’s count for its augmented coupling matrix, reproduced here from an independent 3-D Cartesian derivation. The agreement is a corroboration of the thesis construction, not a restatement of it.

The transfer identity is verified three ways: the augmented matrix is constant (no derivative operators survive); the augmented form reproduces the transferred bilinear; and the untransferred and transferred bilinears differ by an explicitly exhibited lateral divergence $\partial_\alpha P_\alpha$.

6 The test space, derived

6.1 The argument

At this point there are ten augmented slots and, for an affine trial space, nine parameters. The temptation is to choose nine conditions. That temptation must be resisted: a subset chosen by scoring against the arbiter it is later judged by is a fitted parameter wearing a derivation's clothes.

There is no choice to make. $\mathbf{J}_6$ is *antisymmetric*, and (4) applies to $\Delta\mathcal{A}$ as well as to $\mathcal{A}$ (it is linear in the operator), so

$$(\mathbf{J}_6 \Delta\mathcal{A})^t = \Delta\mathcal{A}^t \mathbf{J}_6^t = \Delta\mathcal{A}^t (-\mathbf{J}_6) = \mathbf{J}_6 \Delta\mathcal{A}. \quad (18)$$

$\mathbf{J}_6 \Delta\mathcal{A}$ is a **symmetric** matrix, so

$$b[\phi, \psi] = \phi^t \mathbf{J}_6 \Delta\mathcal{A} \psi \quad (19)$$

is a symmetric bilinear form. A symmetric form admits exactly one natural projection:

$$\boxed{\text{test space} = \mathbf{J}_6 \times \text{trial space}.} \quad (20)$$

Verified, together with the negative control: drop $\mathbf{J}_6$ and the symmetry is lost, both at operator level and in the projected system.

6.2 The ten-versus-nine mismatch dissolves

The ten slots were never test conditions. They measure how many derivative channels the *form* touches — its representation — not how many independent functionals it supplies. Each trial function ψ_k gives exactly one test functional $(\mathbf{J}_6 \psi_k)^t$, so the number of conditions equals the trial dimension by construction.

6.3 The effective coupling matrix

For the affine trial space — nine parameters, three rigid translations and six uniform strains — the projected coupling

$$(\Delta\mathbf{C}_{\text{eff}})_{kl} = \int_V (\mathcal{L} \psi_k)^t \Delta\mathbf{C} (\mathcal{L} \psi_l) dV \quad (21)$$

is the 3-D Cartesian analogue of the thesis's effective coupling matrix. It is 9×9 , symmetric, of full rank 9, vanishes with the contrast, and — because the trial functions are polynomials — requires **no Green's function at all**: the cube integral is exact.

The rigid-translation block is diagonal,

$$(\Delta\mathbf{C}_{\text{eff}})_{1:3,1:3} = i\omega^3 \Delta\rho V \mathbf{I}_3, \quad V = (2a)^3. \quad (22)$$

The power of ω is a basis statement, not a physical one. $\Delta\mathcal{A}$'s density entry is $i\omega\Delta\rho$ acting on the velocity slot; the trial carries $\mathbf{v} = -i\omega\mathbf{u}$ there; and $\mathbf{J}_6$ pulls the test function's

velocity slot into the traction rows, supplying a second $-i\omega$. Hence $i\omega \cdot (-i\omega) \cdot (-i\omega) = i\omega^3$. The displacement-basis answer $i\omega\Delta\rho V$ is wrong by ω^2 . This is the displacement-versus-flux confusion in a new place, and the check asserts the correct factor *and* asserts the incorrect one is incorrect, so it cannot drift back.

7 What the first-order coupling actually buys

7.1 The trial spaces coincide

It is tempting to attribute any difference from the Navier formulation to the nonlinearity of §4.3, or to a different ansatz. Neither is right. Inside the scatterer the moduli are constant, so $\boldsymbol{\tau}_3 = \mathbf{C}_{33}^{\text{tot}} \boldsymbol{\varepsilon}_{i3}$ is a constant *invertible* map. A uniform strain gives a uniform $\boldsymbol{\tau}_3$ and an affine $\mathbf{v}$, and conversely: “ $\boldsymbol{\varepsilon}$ uniform” and “ $\boldsymbol{\tau}_3$ uniform plus in-plane $\boldsymbol{\varepsilon}$ uniform” describe the same nine-parameter family.

The two formulations are therefore **different projections of the same residual onto the same trial space**. The difference lives in the test conditions.

7.2 The density channel cannot differ

With $\Delta\lambda = \Delta\mu = 0$, $\Delta\mathcal{A}$ is exactly $i\omega\Delta\rho$ in rows 1–3, columns 4–6, and reaches *no* part of the stress-glut slot: a density contrast is a pure body force. The propagator’s body-force velocity block is $-i\omega\mathbf{G}$, so both routes return

$$A_u = \frac{1}{1 - \omega^2 \Delta\rho \Gamma_0}, \quad \Gamma_0 = \int_V \mathbf{G} dV, \quad (23)$$

identically. The modulus channels are the only place the two can part.

7.3 Born agreement, and an $O(\Delta c^2)$ difference

Reading a proportionality constant off the rigid block alone gives $i\omega$, and testing it on the remaining eighty entries **refutes** the hypothesis $\Delta\mathbf{C}_{\text{eff}} = i\omega\Delta\mathbf{C}_{\text{Navier}}$. The two couplings agree exactly at $O(\Delta c^0)$ and $O(\Delta c^1)$ — identical Born limits, so nothing is broken — and differ from second order.

Where. Eleven of eighty-one entries differ. With the parameter order $(c_1, c_2, c_3, e_{11}, e_{22}, e_{33}, e_{12}, e_{13}, e_{23})$ the nonzero block is $\{e_{11}, e_{22}, e_{33}\}^2$ together with (e_{13}, e_{13}) and (e_{23}, e_{23}) .

That is exactly the set of strains $\boldsymbol{\tau}_3$ sees:

$$\tau_{13} = 2\mu e_{13}, \quad \tau_{23} = 2\mu e_{23}, \quad \tau_{33} = \lambda(e_{11} + e_{22} + e_{33}) + 2\mu e_{33}. \quad (24)$$

The in-plane shear e_{12} is **untouched**, as it must be: τ_{12} was eliminated, not resummed. This was a falsifiable prediction about which entries may be nonzero, and it held. A representative entry is

$$(\Delta\mathbf{C}_{\text{eff}} - i\omega\Delta\mathbf{C}_{\text{Navier}})_{e_{11}e_{11}} = -\frac{i\omega V \Delta\lambda^2}{\lambda + 2\mu}, \quad (25)$$

pure second order, divided by K_c : the remainder of the geometric series for $1/(K_c + \Delta K_c)$.

7.4 Half of it is a broken symmetry

An isotropic cube in an isotropic whole space has an O_h -symmetric T -matrix; no direction is special. But the first-order system is a recursion in *depth*: it eliminated τ_1, τ_2 and kept τ_3 , so x_3 is preferred by construction. The tell is already above — (e_{13}, e_{13}) and (e_{23}, e_{23}) differ while (e_{12}, e_{12}) does not, and those three are O_h -equivalent.

Applying the cyclic element $x \rightarrow y \rightarrow z \rightarrow x$ (positive controls: $P^3 = \mathbf{I}$, and the Navier coupling *is* invariant):

contrast (GPa)	invariant	breaking	ratio
pure shear, $\Delta\mu = 1$	2.62×10^{-2}	1.94×10^{-2}	1.3520
moderate, (2, 1, 0.1)	5.17×10^{-4}	2.93×10^{-4}	1.7669
strong, (20, 10, 0.5)	1.05×10^{-2}	5.92×10^{-3}	1.7669

Table 1. Decomposition of $\Delta\mathbf{C}_{\text{eff}} - i\omega\Delta\mathbf{C}_{\text{Navier}}$, relative to $\|\Delta\mathbf{C}_{\text{eff}}\|$, at $\lambda = 17.5$, $\mu = 22.5$ GPa. The two mixed rows share a contrast *direction* ($\Delta\lambda:\Delta\mu = 2:1$) at different magnitudes and return the identical ratio, so the split is a function of contrast direction alone — as a leading-order $O(\Delta c^2)$ split must be.

So the difference is neither pure artefact nor pure physics. It carries both a genuine O_h -invariant resummation the Navier form lacks, and a spurious preference for x_3 .

A caveat on this test. The group used is the cyclic subgroup of order 3, not all 48 elements of O_h . Invariance under it is necessary for O_h invariance, not sufficient.

7.5 What follows, and what does not

Two readings are available and the measurement above does not separate them.

Against the route: an approximation that breaks a symmetry the exact answer possesses has introduced an error that no propagator can remove.

For the route: the correction is not scattered. It lands exactly on the traction channel, treating $\mathbf{C}_{33}$ exactly instead of leaving it to the Neumann series, and leaves the in-plane channel alone. Vertical P incidence excites e_{33} ; vertical S incidence excites e_{13} or e_{23} ; both are corrected. The in-plane shear, which the side-scattered field carries, is unchanged. If the response is dominated by reflection and transmission, the asymmetry is “z treated better” rather than “z treated worse”.

Consequence for any comparison. Do not score the raw coupling against an independent arbiter in a homogeneous background. Three effects are in play at comparable magnitude — the Galerkin-versus-collocation projection, the O_h -invariant resummation, and the artefact — and a single number separates none of them. Any comparison must be channel-resolved.

Whether the resummation is an *improvement* is not established here. The local inversion $1/(K_c + \Delta K_c)$ is exact; whether that makes the closed T -matrix better depends on how the ansatz error and the series truncation interact, and those do not simply add.

8 The propagator, and the bridge to it

8.1 It already exists

Everything downstream needs the Green’s matrix of (1). It does not need to be built: the stratified 6×6 Green’s function and its validated wrapper already return it, in the basis $(u_z, u_x, u_y, T_{zz}, T_{xz}, T_{yz})$ — which is exactly the basis the thesis writes $\mathbf{A}$ in. That the two are the same object had, however, been *assumed*.

8.2 The bridge identity

Away from the source plane the Green’s matrix solves the homogeneous system in its receiver coordinate, so for two receiver depths in one uniform layer

$$\Gamma(z_2, z') = \exp(\mathbf{A}(z_2 - z_1)) \Gamma(z_1, z'), \quad (26)$$

with no free parameters. The identity is *local*: it holds however complicated the medium between the source and z_1 is, because the ODE knows only the layer the receiver sits in. The gate therefore places the source above a fast slab and the receivers below it, so transmission, reflection, conversion and interbed multiples are all present in Γ and (26) must still hold exactly.

8.3 Reporting residuals against the arithmetic floor

A bare relative residual is not comparable across incidence angles. Beyond the critical slowness the propagator carries both a growing and a decaying evanescent branch, so $\exp(\mathbf{A} dz)$ has dynamic range $\sim e^{|k|dz}$ — about 1.6×10^7 at the widest angle used. Forming $\exp(\mathbf{A} dz) \Gamma(z_1)$ then amplifies the rounding already present by $\|\exp(\mathbf{A} dz)\|$, and no correct implementation can beat

$$\text{floor} = \epsilon \|\exp(\mathbf{A} dz)\| \frac{\|\Gamma(z_1)\|}{\|\Gamma(z_2)\|}. \quad (27)$$

Residuals below are quoted in units of that floor, so $O(1)$ means the identity holds to the last bit available at that conditioning.

k_x	k_y	$ k /\omega$	raw	corrected
2.00	1.30	0.0759	7.68×10^{13}	1.26×10^{-2}
0.40	0.05	0.0128	2.95×10^{14}	2.43×10^{-2}
14.00	9.00	0.5298	4.31×10^{13}	9.37×10^{-3}
control, $\zeta \leftrightarrow \chi$ swapped			7.68×10^{13}	1.36×10^{12}

Table 2. Residual of (26) in units of the arithmetic floor. $\omega = 31.4159 \text{ rad s}^{-1}$, $dz = 1 \text{ km}$. The third row is evanescent for both branches.

The corrected object is the physical state vector of $\partial_3 \mathbf{q} = \mathcal{A} \mathbf{q}$; the raw one misses by fourteen orders of magnitude.

The negative control swaps ζ and χ — two in-plane stiffnesses that differ by exactly 2μ , a plausible transcription slip that no symmetry check would catch. It is rejected by fourteen orders.

9 Summary of checks

object	what is established	checks
$\mathcal{A}$	built from c_{ijkl} ; matches the published isotropic blocks	4
	matches the thesis $\mathbf{A}(k_x, k_y)$ under one similarity transform	3
	symmetry relations, spectrum, normal incidence	6
	stiffness matrices, $\mathbf{U}_{al}$, ν_1 , ν_2	9
	source vector, negative control	3
$\Delta \mathcal{A}$	exact contrast operator; Navier body force on a generic field	4
	nonlinearity in the contrast, quantified	3
	augmentation: constant, reproduces the form, divergence exhibited	4
	O_h /trial-space structure, density channel	6
$\Delta \mathbf{C}_{\text{eff}}$	$\mathbf{J}_6 \Delta \mathcal{A}$ symmetric; test space derived; controls	5
	9×9 symmetric, rank 9, rigid block, basis trap	7
	Born agreement, $O(\Delta c^2)$ difference, localisation	9
	O_h decomposition, positive and negative controls	6
Γ	propagates by $\exp(\mathbf{A} dz)$ with all multiples present	3

Table 3. All checks pass. The two symbolic scripts report 25/25 and 44/44; the propagator gate exits clean.

10 What is not established

In keeping with the project’s standard that an unvalidated formula be labelled as such, the following are *open*:

1. **No single-site T -matrix has been assembled.** $\Delta \mathbf{C}_{\text{eff}}$ is the coupling; the propagator moments $\langle \mathbf{J}_6 \psi_k, \Gamma \psi_l \rangle$ that close $T = \Delta \mathbf{C}_{\text{eff}} (\mathbf{I} - \mathcal{G} \Delta \mathbf{C}_{\text{eff}})^{-1}$ have not been computed.
2. **No comparison against an independent arbiter has been run.** The quantities in §7 compare two *formulations* with each other, not either against truth. Section 7.4 argues such a comparison must be channel-resolved before it can mean anything.
3. **Whether the resummation improves accuracy is unknown.** It is exact as a local inversion; that is not the same claim.
4. O_h **invariance was tested on a subgroup of order 3**, not the full group.

5. **The derivative count for a collocation scheme differs from the Galerkin one.** In the Galerkin pairing the transferred derivative lands on polynomials, so the third-derivative moment cannot arise; under collocation it transfers onto $\mathbf{\Gamma}$ and can. Do not quote the Galerkin count for a collocation build.

11 Where the route belongs

The x_3 preference of §7.4 is an artefact only because the problem posed there — an isotropic cube in an isotropic whole space — has no preferred direction. A layered background does. There the formulation’s directional structure matches the physics rather than contradicting it, the O_h argument does not apply, and the propagator is already available and already bridged to $\mathcal{A}$ by (26).

That is also the case in which the ordinary route has no competitor: the cube moment method rests on volume moments of the whole-space Green’s tensor, and in a stratified background that object does not exist. The first-order form supplies what the moment method cannot.